DIIT
Daffodil Institute of IT

[SET: Loop]

DAFFODIL INSTITUTE OF IT

B.Sc (Hons) in Computer Science & Engineering

Midterm Examination of CSE 1st Year 1st Semester, 24th Batch, 2025

510201 (Structured Programming Language)

Time – 1 hour 20 minutes, Full marks – 40

[N.B. – The figures in the right margin indicate full marks. Answer any ***two*** *questions. All parts of the questions must be answered sequentially.]*

			Marks
1.	(a)	What is structured programming? Write down the importance of C programming language.	1+4=5
	(b)	Explain different control structures used in structured programming.	6
	(c)	What do you mean by preprocessor directive? Why and when do we use the #include and #define directives? Explain with example.	1+4=5
	(d)	Write down the algorithm to calculate area and circumference of a circle.	4
2.	(a)	Define token, keyword, identifier, variable and constant with example	1+4=5
	(b)	What is variable? How can you define and initialize a variable in C program?	1+4=5
	(c)	What is operator? With example describe the rules for ++ and -- operator.	1+4=5
	(d)	Write a C program to verify whether a given year is a leap year or not.	5
3.	(a)	Describe the following statements with general form and flowchart: (i) nested if…else (ii) for	3+3=6
	(b)	What is infinite loop? Explain entry and exit control loop.	1+4=5
	(c)	Mention the differences between while and do while loop.	4
	(d)	Write a program in C to determine and print the sum of the following harmonic series for a given value of n: $s = 1 + 2^2 + 3^3 + \cdots + n^n$	5

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			Marks
1.	(a)	Define structured programming language. Write down some salient features of structured programming language.	1+4=5
	(b)	Describe the basic structure of a C program.	6
	(c)	What is pseudo code? Distinguish between algorithm and flowchart.	1+4=5
	(d)	Draw the flowchart to calculate the temperature from Celsius to Fahrenheit.	4
2.	(a)	Describe with example the four basic data types used in a C program.	5
	(b)	What is symbolic constant? Write down the rules for declaring symbolic constant.	1+4=5
	(c)	Define operator and expression. Distinguish between p++ and ++p.	2+3=5
	(d)	Write a C program to determine the smallest value among three numbers.	5
3.	(a)	Describe the following statements with general form and flowchart: (i) if…else (ii) while	3+3=6
	(b)	What is infinite loop? Explain entry and exit control loop.	1+4=5
	(c)	Mention the differences between break and continue statements.	4
	(d)	Write a C program to determine and print the sum of the following series for a given input value of N: $s = 1 + 2^3 + 3^3 + \cdots + N^3$	5